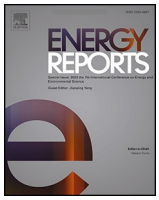
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Research paper



Comparison of virtual energy sharing schemes for renewable energy communities under European regulations: A mixed-use case study in Italy

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ABSTRACT

The Renewable Energy Directive (RED II) introduced Renewable Energy Communities (RECs) as a mechanism to empower local actors in the energy transition through collective renewable energy sharing. European Member States have transposed RED II into diverse regulatory frameworks and incentive schemes, leading to different approaches to electricity sharing. This study analyses the techno-economic performance of a mixed-use REC composed of a university campus and a residential district, using City Energy Analyst to simulate six scenarios combining two virtual energy sharing schemes, Virtual Self-Consumption (VSC) and Virtual Net Metering (VNM), and three ownership configurations, ranging from a centralized external producer to a distribution of prosumers. Sensitivity analyses on PV sizing and electricity market prices are also performed. Results show that VSC schemes achieve higher energy performance than static VNM schemes due to more effective matching between generation and demand, while economic outcomes remain comparable across schemes. Compared to the baseline, energy sharing increases self-consumption by up to 35% in VSC schemes and up to 25% in VNM schemes, with revenue increases of up to 20% under both schemes. Configurations involving non-residential prosumers improve self-consumption and enable higher PV penetration without a proportional loss in economic viability. At the same time, economic performance is largely driven by accounting rules, including incentive tariffs and market price structures, which can offset differences in energy performance between sharing schemes. These findings suggest that REC design should prioritise the integration of non-residential prosumers, while regulatory frameworks should align economic incentives with energy efficiency objectives.

1. Introduction

The decarbonisation of the energy sector has become a central objective of international and European climate policy, as the rapid reduction of greenhouse gas emissions is required to meet the targets of the Paris Agreement and the European Green Deal. (UNFCCC, 2015; European Commission, 2019a). In this context, the large-scale deployment of renewable energy sources (RES), particularly distributed photovoltaic (PV) generation, is a cornerstone of the energy transition toward more decentralised and low-carbon configurations. However, increasing shares of distributed generation also introduce new technical, economic, and governance challenges for electricity system operation, distribution grid management, and social acceptance of infrastructure development (Interngovernmental Panel on Climate Change IPCC, 2011). To address these issues, European policy has progressively shifted from a purely market-based approach toward models that actively involve end-users as participants in energy production and management. The Clean Energy for All Europeans package (CEP) (European Commission, 2019b), through the Renewable Energy Directive (RED II) (European Union, 2018) and the Internal Market for Electricity Directive (IEMD) (European Union, 2019), formally recognizes Renewable Energy Communities (RECs) as instruments to enable collective investment in renewable energy, promote local energy use, and deliver social, environmental, and economic benefits at the community level. While EU legislation establishes the right of RECs to share renewable electricity among their members, it leaves substantial discretion to Member States regarding the regulatory design and implementation of energy sharing mechanisms. As a result, a heterogeneous landscape of enabling frameworks has emerged from national policy transpositions, differing in accounting principles, incentive structures, and operational requirements for electricity sharing (Energy Community Repository, 2023a; RESCOOP, 2024). These differences are particularly evident for virtual electricity sharing, which relies on administrative and

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Nomenclature

AB	Apartment Block
BEM	Building Energy Model
CEA	City Energy Analyst
CEC	Citizen Energy Community
CEP	Clean Energy Package
DER	Distributed Energy Resources
DSO	Distribution System Operator
EU	European Union
FiP	Feed-In Premium
FiT	Feed-In Tariff
GFA	Gross Floor Area
HDD	Heating Degree Days
IEMD	Internal Electricity Market Directive
LEM	Local Energy Market
MFH	Multi-Family House
P2P	Peer-to-Peer
POD	Points-of-Delivery
PPA	Power Purchase Agreement
PV	Photovoltaic
REC	Renewable Energy Community
RED II	Renewable Energy Directive
RES	Renewable Energy Sources
SME	Small and Medium-sized Enterprises
TIP	Italian Incentivized Premium Tariff
TMY	Typical Meteorological Year
UBEM	Urban Building Energy Model
UNIPA	University of Palermo
VNB	Virtual Net Billing
VNM	Virtual Net Metering
VSC	Virtual Self-Consumption

accounting mechanisms rather than physical self-consumption. Among the most common schemes adopted across European countries are Virtual Net Metering (VNM), which allows renewable production to be virtually allocated among members over a given accounting period, and Virtual Self-Consumption (VSC), which incentivises the simultaneous consumption of locally generated electricity within the community. These mechanisms differ substantially in the way shared electricity is calculated and remunerated, potentially leading to different techno-economic outcomes for the same technical configuration. Several contributions have examined the policy design and institutional implications of different energy sharing mechanisms for the development of RECs and collective self-consumption schemes (Gjorgievski et al., 2023; Llera-Sastresa et al., 2023; Minuto and Lanzini, 2022; Taromboli et al., 2025). At the same time, a growing body of literature focuses on the optimization of community energy systems, typically addressing PV sizing, energy sharing strategies, or benefit allocation among participants (Belloni et al., 2024; Buonomano et al., 2025; Calise et al., 2024; Mansó Borràs et al., 2023). However, these approaches often rely on simplified representations of either regulatory mechanisms or building-level energy dynamics. As a result, the extent to which different enabling frameworks influence the actual techno-economic performance of RECs remains insufficiently quantified. This study contributes to bridging this gap by combining a detailed Urban Building Energy Model (UBEM) of community energy demand with the comparative evaluation of alternative schemes for virtual electricity sharing. Beyond regulatory design, the ownership structure of renewable energy assets represents another key dimension of RECs. Generation assets may be owned collectively through cooperative arrangements, by public institutions, or by private actors participating in the community. These ownership models can differ in the degree of centralisation: renewable generation may be concentrated in a few large installations controlled by a single actor, or distributed across multiple smaller systems owned by individual prosumers. Such differences affect not only the allocation of investment costs and revenues, but also the governance of the community and the mechanisms through which benefits are distributed among participants.

2. Literature review

2.1. Renewable energy communities

The Clean Energy for All Europeans (CEP) legislative package, which includes the Renewable Energy Directive (RED II) and the Internal Market for Electricity Directive (IEMD), introduced measures to support local communities in the energy transition by formally recognizing new participants in the energy market, notably Renewable Energy Communities (RECs) and Citizen Energy Communities (CECs). This framework aims to empower consumers across Europe by highlighting the added value ECs bring by enhancing social acceptance of renewable energy, encouraging local engagement and investment, expanding consumer choice, fostering greater public participation, and prioritizing affordable, community-based energy solutions (Energy Community Repository, 2024). Although RECs and CECs differ in scope, both are defined as legal entities operating under national law, based on open, voluntary membership, and controlled by members or shareholders to deliver community-focused energy solutions. Both are grounded in the principle of uniting citizens, small and medium-sized enterprises (SMEs), and local authorities under democratic governance to promote local ownership of sustainable energy projects and provide social, environmental, and economic benefits to the community over making profits. While Citizen Energy Communities may have a broad geographic scope and include non-renewable energy in their operations, RECs have an inherently local focus. Article 22(2) of RED II outlines the entitlements of RECs, which include producing, consuming, storing, and selling renewable energy; and sharing this energy among members, while preserving their rights as energy customers (European Union, 2018).

2.2. Energy sharing schemes

While several countries have yet to introduce regulation on electricity sharing, most Member States have developed enabling frameworks to operationalize the provisions of the RED II, resulting in a diverse range of regulatory models for electricity sharing within RECs (Gjorgievski et al., 2023; Llera-Sastresa et al., 2023; Energy Community Repository, 2023b). These models differ primarily in the allocation of shared energy, the role of distribution system operators (DSOs), the structure of tariff incentives, and the degree of administrative integration. A first important distinction can be made between *direct energy sharing*, for schemes based on physical collective self-consumption, and *virtual energy sharing*, for schemes based on accounting of electricity production and consumption without physical self-consumption. Among these, two main approaches have emerged as the prevailing paradigms for implementing virtual electricity sharing across Europe: *Virtual Self-Consumption (VSC)*, based on the ex-post accounting of the simultaneity between electricity production and consumption, and *Virtual Net-Metering (VNM)*, based on an ex-ante static or dynamic allocation of PV production shares. Beyond these schemes, some jurisdictions are also experimenting with Local Energy Markets (LEM) and Peer-to-Peer (P2P) trading models (Dudjak et al., 2021; Bukar et al., 2023), which are based on the establishment of internal electricity prices and local transactions among participants.

2.2.1. Direct energy sharing

Physical or direct energy sharing refers to schemes based on collective self-consumption enabled by a direct electrical connection between one or more renewable generation units and the consumption points of community members. This configuration is commonly referred to as behind-the-meter energy sharing, as the electricity is physically delivered through a private network or behind a single connection point to the public grid, allowing locally produced renewable energy to be consumed on-site without using the public distribution network. Accordingly, the allocation of energy among participants follows the actual physical flows, typically leading to higher self-consumption rates and lower effective electricity costs compared to individual self-consumption, as locally shared renewable electricity limits grid imports and reduces exposure to network charges and levies. In turn, physical sharing configurations are limited to proximate consumers tied to the same connection point and need dedicated infrastructural investments and technical adaptations, including private cabling and metering, which can increase costs and limit feasibility, especially in urban settings with space constraints and complex permitting procedures (Ball and Cappers, 2026).

2.2.2. Virtual energy sharing

Virtual energy sharing schemes decouple electricity sharing from physical power flows, enabling members of an Energy Community to share the benefits of renewable generation even if they are not directly connected to the points of production. Under this arrangement, the electricity grid effectively functions as a "virtual battery", allowing renewable energy produced at distributed sites to be accounted for and allocated among community members. The IEMD mentions that *"CECs should not face restrictions to share electricity […], for example by offsetting the energy component of members using the generation available within the community, even over the public network, provided that both metering points belong to the community"* (European Union, 2019). This technical design relies on metering and accounting mechanisms that track generation and consumption separately, enabling members to receive credits or financial incentives for renewable production they are entitled to. The grid operator (or a balancing party) acts as an intermediary, tracking production and consumption, applying the sharing key to allocate production, and settling the corresponding credits or incentives, ensuring transparent and accurate distribution of the shared electricity. This approach provides access to renewable electricity without ownership of a generation system, expanding participation to consumers lacking suitable rooftop, house-ownership, or investment capacity. Members can benefit economically through bill reductions, incentives, or virtual allocation of generated energy, while maintaining the flexibility to consume electricity according to their demand patterns. From the grid perspective, virtual energy sharing can help balance supply and demand, reduce peak loads, and mitigate local congestion risks by aggregating distributed generation across multiple locations. Overall, virtual energy sharing facilitates higher penetration of renewable energy by enabling flexible integration of distributed energy resources (DERs). Fina et al. (2019) show that the profitability of PV systems increases when forming RECs with multiple buildings compared to energy sharing in individual buildings. Minuto and Lanzini (2022) identify two main energy sharing schemes proposed in the literature: peer-to-peer (P2P), in which self-produced energy is traded among different points-of-delivery (PODs), and virtual metering, in which the net energy flux exchanged between different PODs is as if they belong to the same virtual POD. They also describe methods in which energy procurement can be accomplished in a REC: Power Purchase Agreements (PPA); Feed-In Tariffs (FiT) and Premiums (FiP) on the energy generated and fed into the grid; Virtual Net-Metering (VNM), in which the energy withdrawn from the grid is reduced by the amount fed into the grid within a certain time frame; and Virtual Self-Consumption (VSC), in which the withdrawn energy is paid on a retail basis, while the self-consumed energy is incentivized. Moreover, Gjorgievski et al. (2022) propose a virtual net-billing (VNB) method by applying dynamic sharing keys to a VNM logic. Based on the previous classifications of energy sharing mechanisms (Minuto and Lanzini, 2022; Bukar et al., 2023; Gjorgievski et al., 2022), this study proposes the following operative definition to represent the majority of the sharing schemes adopted by European countries:

- Virtual Self-Consumption (VSC): a model where renewable energy produced by a community plant is allocated to members based on the consumption occurring simultaneously with the shared renewable production; members receive incentives or bill reductions for the energy accounted for. The VSC model relies on the active role of the DSO or another balancing entity to calculate the volume of shared electricity at specific intervals (typically 15 min or one hour). The amounts are invoiced or communicated to the responsible party (e.g. suppliers) to reward users who consume during periods of renewable generation. Reduced network tariffs or exemptions from grid fees can be applied to the portion of electricity that qualifies as shared energy.
- Virtual Net-Metering (VNM): a model where renewable energy produced by a community plant is allocated to members according to fixed or variable shares, regardless of when members consume electricity; the allocated shares are credited on members' bills, reducing net consumption without requiring time-based matching between generation and use. The VNM model relies less on DSOs for continuous data exchange, since allocation does not depend on real-time consumption. Instead, it adopts a simplified accounting logic, though the administrative management falls more on the RECs themselves.

2.3. Enabling frameworks for energy sharing

EU countries were bound to transpose into their national legislation the definitions for RECs and CECs, as well as enabling frameworks and national support schemes for RECs introduced by the RED II and IEMD. The RESCOOP Transposition Tracker monitors the transposition process and the alignment of national regulations to EU Directives in terms of definitions and enabling frameworks. Different studies offer a review of policies and supporting mechanisms for RECs implemented by European countries (Gjorgievski et al., 2023; Llera-Sastresa et al., 2023; Minuto and Lanzini, 2022; Taromboli et al., 2025) or a detailed description of national implementation frameworks (Couraud et al., 2025; Fina and Fechner, 2021; Fischer et al., 2024; Kortetmäki et al., 2024). In 2023, the Energy Community Repository provided an overview of the enabling frameworks and the energy sharing schemes set up by all EU countries (Energy Community Repository, 2023b and 2024). A classification of the energy sharing schemes adopted is reported in Table 1 and in Appendix A.1.

Most countries have implemented virtual energy sharing schemes that could be categorized as Virtual Self-Consumption (VSC) and Virtual Net-Metering (VNM). Some have implemented LEM (e.g. Denmark, France) or P2P trading (e.g. Flanders region). These frameworks differ mainly in how shared energy is allocated, the role of DSOs, the structure of tariff incentives, and administrative arrangements. In countries with VNM-like schemes (e.g. Austria, Croatia, Spain), the DSO is responsible for computing the balance between the energy consumed and the energy generated allocated to the participants at regular intervals (15 min or an hour). The allocation follows a sharing key (static or dynamic) defined by the REC in accordance with the DSO. For billing purposes, members' suppliers are informed by the REC or by members about the sharing key adopted (e.g. Slovenia, Spain), or by the DSO about the allocated amounts (e.g. Austria, Finland). REC members often benefit from reduced network tariffs or tax exemptions on the shared energy (e.g. France, Hungary, Spain) or on total consumption (e.g. Brussels region, Croatia). Surplus energy can generally be sold to a third-party supplier, chosen collectively (e.g. Luxembourg) or individually (e.g. France), or to a responsible party (e.g. Italy). Gjorgievski et al. [9] detected a significant variability in the impact of regulated charges and exemptions for

Table 1
Energy sharing incentive schemes are adopted by European countries.

Incentive	**Energy sharing**	**FiP/ FiT**	Fiscal	LEM/P2P	Country reference
Austria	VNM	-	✓	-	(Fina and Fechner, 2021; Fischer et al., 2024)
Belgium	VNM (only Wallonia)	-	✓ (only Brussels)	✓ (only Flanders)	-
Croatia	VNM	-	✓	-	-
Denmark	-	-	-	✓	-
Finland	VNM	-	-	-	(Kortetmäki et al., 2024)
France	-	-	✓	✓	(Couraud et al., 2025)
Germany	VSC	✓	✓	-	-
Greece	VNM	-	-	-	-
Hungary	-	-	✓	-	-
Ireland	-	✓	-	-	-
Italy	VSC	✓	-	-	(Lage et al., 2024)
Luxembourg	VNM	-	✓	-	-
Netherlands	-	✓	-	-	-
Portugal	VNM	-	✓	-	(Lage et al., 2024)
Slovenia	VNM	-	-	-	-
Spain	VNM	✓	-	-	-
United Kingdom	-	✓	-	-	-

VNM: Virtual Net-Metering // ***VSC:*** Virtual Self-Consumption
FiP/FiT: Feed-in Premium, Feed-in Tariff, Net purchase and sale, Subsidy on generated energy
Fiscal: Tax exemptions, Reduced grid tariffs // ***LEM/P2P:*** Local Energy Market, Peer-to-Peer Trading

shared energy in European countries. As for 2023, rules for electricity sharing were under development in Latvia, Malta, and Ireland, despite having already partly transposed CEP directives. Bulgaria, Cyprus, Poland, and Slovakia had not yet implemented any enabling framework.

2.4. Ownership of energy assets

RECs can adopt different configurations regarding the ownership of energy generation assets, which affect not only the distribution of economic benefits but also the operational logic of electricity sharing and the incentives for participation (Dudka et al., 2023). In practice, renewable energy (RES) plants are generally owned by the REC itself or by its members, acting as prosumers that share part of their own generation with the community or as investors owning a quota of a larger collective plant. RECs could have different degrees of centralization, opting for multiple small-scale plants or few large installations. The aggregation of members within RECs also affects their management and performance, and could be controlled according to indicators linked to users' consumption (Buonomano et al., 2026). Often, a collective plant is owned by a public actor, such as the municipality, or a third-sector entity that initiated the REC project. Key decisions regarding asset management, energy allocation, and the distribution of economic benefits are taken democratically by the governing body of the REC. Some arrangements may be constrained by the regulatory frameworks, such as limits on profit distribution and on participation of big market actors imposed by the European RED II. Still, national regulations can be more permissive and allow the participation of market-oriented players under specific configurations; in Italy, for example, third-party producers can provide their renewable generation to a REC to enable the generation of incentives for virtual energy sharing, even when they cannot legally become a member. Fig. 1 depicts some of these configurations.

2.5. REC modelling

RECs have been modelled with different objectives: the planning and design of REC composition, the sizing of RES and storage systems, the optimization of energy and economic indicators, the feasibility of technical configurations, and so on. Vecchi et al. (2024a) reviewed the tools available for REC modelling, focusing mostly on Building Energy Models (BEM). Although no exhaustive REC modelling tool was identified, the review suggests either making advantage of the complementarity of different specialised tools, for example in a co-simulation framework, or performing a broader analysis using tools able to process a comprehensive amount of spatialized data but with lower accuracy, such as City Energy Analyst (CEA) and URBANopt. Some studies developed tailored co-simulation frameworks to couple BEMs, mostly run through EnergyPlus, with other components used to simulate power systems and users' behaviour. For example, Brandi et al. (2025) developed a co-simulation architecture using the Python-based framework Mosaik, coupled with EnergyPlus and a Modelica model for heating systems, to compare REC scenarios for 211 buildings in a small Italian municipality. Giuzio et al. (2024) modelled a REC composed of three (3) buildings to investigate user demand flexibility, using a co-simulation platform in the MATLAB/Simulink environment that integrates EnergyPlus into a District Energy System (DES) model. Finally, Belloni et al. (2024) integrated results from EnergyPlus and PVGIS into the EnergyCommunity.jl tool, programmed in Julia, to estimate the thermo-electric needs of a ten-building REC. Other studies relied on

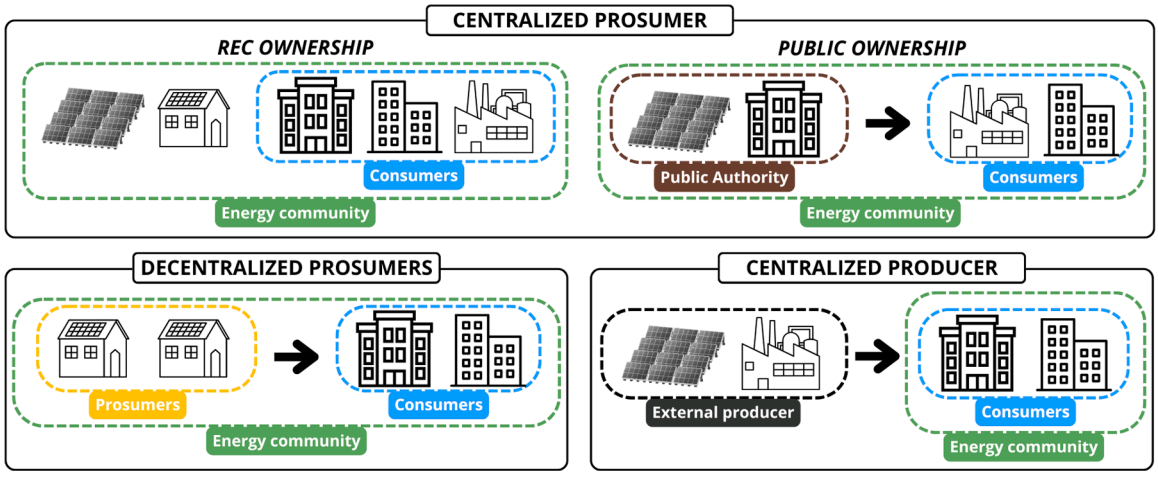


Fig. 1. Common PV asset ownership configurations used by Energy Communities.

UBEM to simulate the energy consumption profiles of the buildings included in their REC case study. For example, Calise et al. (2024) applied the REC model to a residential district of 60 buildings in Naples (Italy), by categorizing them into four (4) archetypes modelled in TRNSYS to simulate the electricity and heating demands. The proposed REC layouts introduced energy sharing at the building and the district level, leading to an increase in self-consumption rates to 32% and 42%, respectively, compared to the baseline of 22% corresponding to individual self-consumption. Furthermore, the district-level REC delivered a 15% reduction in electricity expenses. Tarpani et al. (2025) used the web-platform CityBES, based on the CityGML standard and EnergyPlus BEM, to assess the feasibility of RECs in two areas with around 1300 buildings each, including the historic centre of Perugia (Italy). Buonomano et al. (2025) developed a GIS-based tool running on EnergyPlus to analyse different logics for user aggregation, comparing the implementation of a few large-scale RECs versus many small-scale RECs for a district of 731 buildings in Naples (Italy). Lastly, Mansó Borràs et al. (2023) used CEA to simulate consumption profiles and PVGIS to assess the PV generation potential of a REC composed of 30 buildings in Lisbon (Portugal), and Vecchi et al. (2024b) used CEA to model the consumption and PV production potential of a seven-building cluster, including a university facility, finding that energy sharing increases revenues by 16% compared to individual self-consumption. Overall, the literature highlights that UBEM approaches are particularly promising for REC modelling at the district scale, as they enable the representation of spatial and typological heterogeneity across large building stocks while supporting comprehensive analyses. However, a relevant research gap persists in the analysis of interactions between residential and non-residential buildings within mixed-use RECs. Non-residential buildings often provide larger rooftop areas suitable for PV installation and exhibit electricity demand profiles that are complementary to residential ones, thus offering significant potential for enhanced energy sharing. While some studies explore REC designs combining residential and non-residential users to optimise energy performance, many devote limited attention to how the produced energy is allocated and shared among community members. Suitable non-residential buildings include both public (e.g., schools, hospitals, and sports centres) and private facilities (e.g., commercial and industrial), which can help improve REC performance.

2.6. Knowledge gaps and contribution of the paper

The literature on REC modelling discussed in the previous section is summarized in Table 2. The studies are compared in terms of their scope, modelling approach and tools, energy-sharing schemes, and asset ownership models. Several studies simulate RECs using BEM frameworks, while other studies focus on energy sharing and REC optimization. The combined assessment of detailed community energy dynamics, ownership structures, and policy-based energy sharing schemes remains insufficiently explored. Particularly, the role of non-residential actors in mixed-use RECs as large-scale prosumers and their impact on both energy efficiency and PV deployment strategies is still underexplored in the literature. Indeed, most of the studies presented in Table 2 assume a high degree of decentralization where most buildings act as prosumers, even when non-residential buildings can represent a substantial share of the available PV generation (Vecchi et al., 2024b; Belloni et al., 2024). Instead, the present study clearly compares alternative ownership configurations suitable for the case study, including centralized and decentralized prosumers and an external producer.

As seen in Table 2, the high number of Italian case studies contributes to a prevalence of VSC schemes, reflecting the regulatory framework adopted in the Italian context. While maintaining Italy as the case-study context, the present study stands out from this literature by comparing VSC and VNM schemes. Moreover, limited attention has been given to how different accounting mechanisms may alter the relationship between energy performance and economic outcomes, potentially leading to divergent incentives for REC participants.

This study addresses the research gaps presented in this section by:

- Integrating alternative ownership structures within a REC simulation framework to analyse their influence on the collective energy and economic performance as well as on the distribution of costs and benefits among community members. In this study, three (3) representative ownership configurations are considered: 1) a centralized PV plant owned by an external producer, in this case the university, who installs and operates the system while community members benefit indirectly through energy sharing; 2) a centralized PV plant owned by a REC member acting as a centralized prosumer, participating in energy production and consumption; 3) a set of decentralized PV plants owned by a community of prosumers, all REC members.
- Comparing two regulatory paradigms for virtual electricity sharing, Virtual Self-Consumption (VSC) and Virtual Net Metering (VNM), representing the majority of energy sharing schemes adopted by the policy frameworks of European countries;
- Embedding these elements within a UBEM framework, providing a modelling approach that links detailed building energy dynamics with ownership configurations and policy-based electricity sharing schemes in a mixed-use REC.

3. Method

This study adopts an integrated methodological approach combining

Table 2
Comparison of literature on REC modelling.

Reference	*Scope*	*Modelling method/tool*	*Energy sharing scheme*	*Asset ownership model*
Mansó Borràs et al. (2023)	Residential REC (30 buildings)	CEA + PVGIS	VNM + storage	Decentralized prosumers
Giuzio et al. (2024)	Fictional REC (3 buildings)	MATLAB/Simulink + EnergyPlus + FMU + DES	VSC + flexibility	Not specified
Belloni et al. (2024)	Mixed-use REC (10 buildings)	EnergyPlus + PVGIS + EnergyCommunity.jl	VSC	Decentralized prosumers
Calise et al. (2024)	Residential REC (60 buildings)	TRNSYS	VSC + central HP + storage	Decentralized prosumers
Vecchi et al. (2024b)	Mixed-use REC (7 buildings)	CEA	VSC	Decentralized prosumers
Tarpani et al. (2025)	Urban districts	CityBES + CityGML + EnergyPlus	VSC	Not specified
Brandi et al. (2025)	Municipal-scale REC (211 buildings)	Python Mosaik + EnergyPlus + Modelica	VSC	Decentralized prosumers
Buonomano et al. (2026)	District-scale REC (730 buildings)	GIS + SimStock + EnergyPlus	VSC	Decentralized prosumers

Urban Building Energy Modelling (UBEM) with policy-based energy sharing schemes and alternative ownership configurations to evaluate the techno-economic performance of a mixed-use Renewable Energy Community (REC). The methodology begins with the data collection (3.1 Data collection), in which the case study buildings are characterized through their geometries, thermal and construction characteristics, and available monitored energy consumption. In the modelling phase (3.2 Urban Building Energy Modelling), hourly electricity and heating demand profiles, as well as PV production potential, were simulated for all buildings using the software City Energy Analyst (CEA), based on reference use-types defined according to technical norms. In the REC design phase (3.3 REC design), six (6) scenarios are defined, combining two (2) energy sharing schemes (VSC and VNM) and three (3) PV ownership configurations (centralized producer, centralized prosumer, and decentralized prosumers), allowing a systematic comparison of energy and economic outcomes under consistent technical assumptions. In the data analysis (3.4 Data Analysis), energy performance and economic indicators are calculated to compare results of different scenarios. Also, two sensitivity analyses are performed based on the PV penetration and the yearly electricity prices. The methodological steps described are illustrated in Fig. 2.

3.1. Data collection

The case study consists of 38 buildings located in two districts in Palermo (Italy): 13 on a University campus and 25 in a residential area. Fig. 3 shows the location of the districts within the city, as well as the limit of the primary substation area (orange line) to which they belong. The University campus "Viale delle Scienze" (in yellow) is the biggest campus of the University of Palermo (UNIPA), with a total student population of 44,400 students (around 30,000 estimated in this campus) and 4300 employees (Ministero dell'Università e della Ricerca MUR, 2024). The residential district Cuba-Calatafimi (in blue) is composed of 25 multi-storey apartment blocks built around the 1960–70 s, with a population of 2760 inhabitants (considering an average of 38 m^2 per person). The geometry, the thermal properties of the envelope, and the monitored monthly consumption of electricity and natural gas were available for 13 university buildings, as presented in Table B.1 of Appendix B.

The thermal properties of the buildings in Cuba-Calatafimi were obtained from the Energy Map produced by Palermo municipality (Comune di Palermo., 1998), as reported in Table 3. According to statistical data (ENEA Italian National Agency for New Technologies, Energy and Sustainable Economic Development, 2026; Regione Siciliana, n.d.) and a field visit, no significant modification to the quality of the envelope is expected. The geometry of the residential buildings was obtained directly from the simulation software, after having verified the number of floors in Google Maps. The heating demand of residential buildings was calibrated against two representative TABULA archetypes: a multi-family house (MFH.05) and an apartment block (AB.05) built in Italy around 1960–70 s (TABULA Project., 2016).

3.2. Urban building energy modelling

The UBEM simulation software City Energy Analyst (CEA) was used to obtain yearly profiles of electricity and natural gas consumption, and of PV production potential for each building simulated. These outputs were obtained at the building level with an hourly timestep. The weather input file corresponds to a Typical Meteorological Year (TMY) of the closest urban station of "Palermo Boccadifalco" EnergyPlus, 2026). Detailed information on the simulation parameters used can be found in Appendix B. In particular, Tables B.2 and B.3 detail the main characterization parameters of the simulated university and residential buildings, respectively, including the number of floors, the total gross floor area (GFA), and the estimated occupation. The tables also show the nominal peak power potential under the assumption that half of the available rooftop surface was used for PV production. Table B.3 also reports the heating and cooling systems assigned to each building according to a realistic distribution of technologies adopted in the local context.

Five (5) use-types were defined in CEA to represent the activities carried out in the university buildings: University, Laboratory, Office, Library and Server room. Table 4 shows the main parameters selected for each use-type according to Italian norms.

As shown in Table B.6, each building was assigned up to three (3) use-types according to the available information and the fit to monitored data. Input parameters such as the fraction of conditioned area, the windows-to-wall ratio, the temperature set-points, and the share of electric heating were adapted to fit the monitored data. For most of the university buildings (9 out of 13), the deviation from the monitored consumption is below 5% for either electricity or natural gas (see Fig. 4). Fig. 5 compares the simulated heating demand of residential buildings

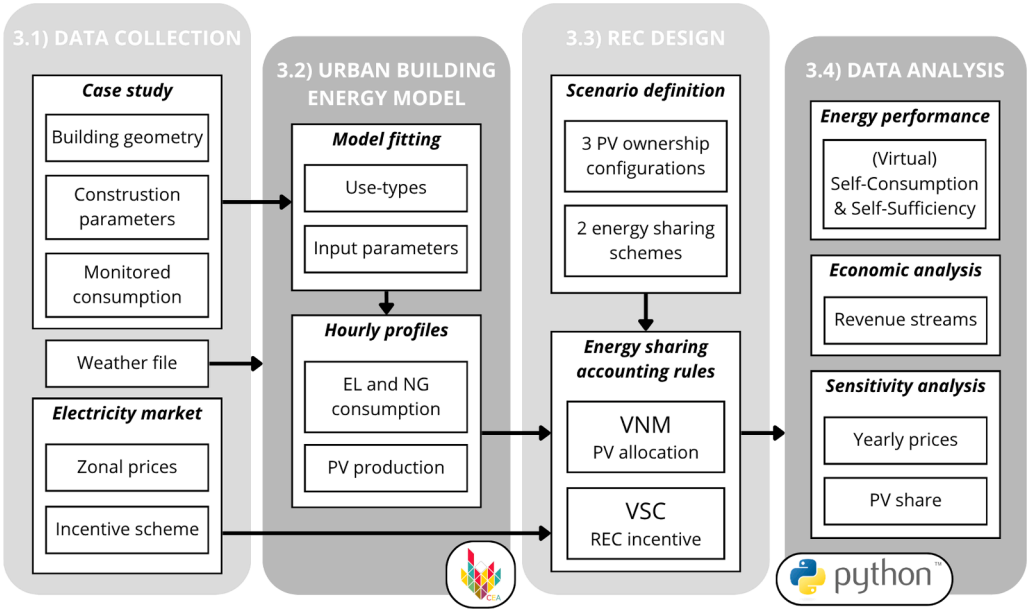


Fig. 2. Graphical representation of the methodological steps implemented.

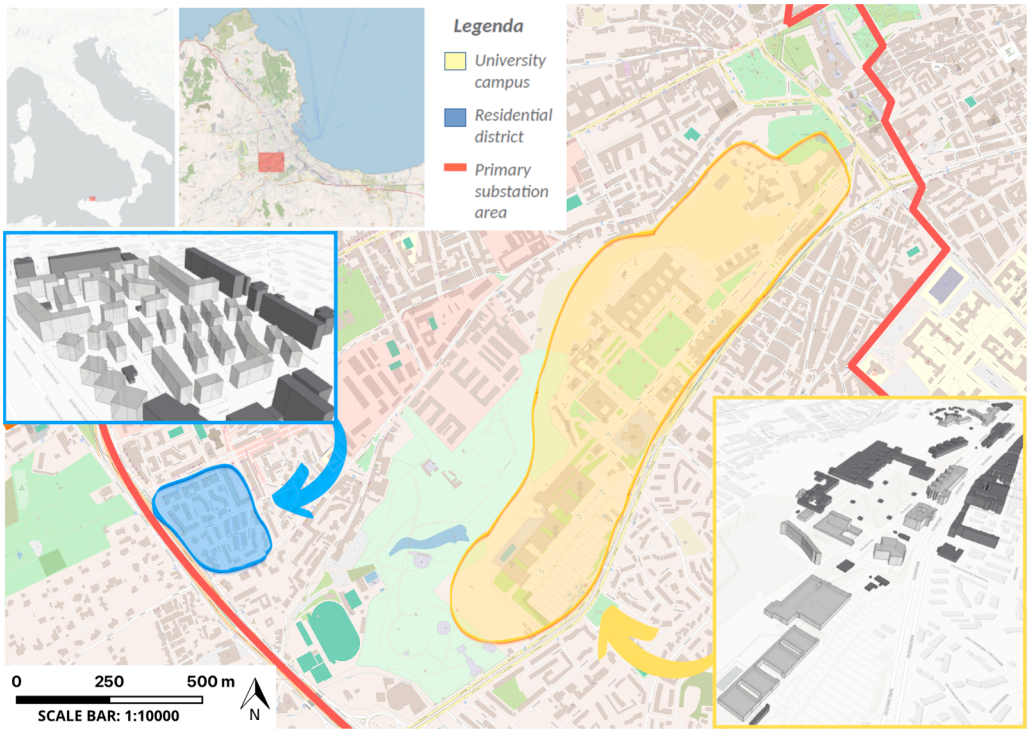


Fig. 3. Location of the case study and outlook of the simulated buildings in CEA.

Table 3
Thermal properties of construction materials in Cuba-Calatafimi.

Envelope component	*Roof*	*External wall*	*Floors*	*Windows*
Construction materials	Brick-cement	Tuff blocks (1.500 kg/m^3)	Brick-cement	Metal frame single glazing
Thickness	25 cm	40 cm	25 cm	3 cm
Transmittance	1.7 W/m^{2}K	1.25 W/m^{2}K	1.67 W/m^{2}K	6.28 W/m^{2}K

against TABULA references, normalized for the Heating Degree Days (HDD) in Palermo.

3.3. REC design

Based on the building-level electricity demand and PV production profiles obtained from the UBEM, six (6) scenarios were elaborated for the same REC, combining two (2) energy sharing schemes (VSC and VNM) and three (3) PV ownership configurations (with different production and consumption profiles), as illustrated in Fig. 6:

- **VSC1: VSC + Centralized producer –** In this scenario, the ***university acts as an external producer*** that only provides its available rooftop area for the installation of PV panels (2.75 MW of nominal power, annual production of 3.96 GWh/y), the electricity produced is sold to the grid and accounted for in the calculation of the incentivized virtual self-consumption, in case any residential building consumes it simultaneously (the peak load is 2.8 MWh and the annual consumption is 9.18 GWh/y);
- **VSC2: VSC + Centralized prosumer –** In this scenario, the ***university acts as a prosumer*** that owns the PV plant on the rooftop of its buildings (same size as VSC1), the electricity produced is first self-consumed by each university building, then sold to the grid and accounted for the calculation of the incentivized virtual self-consumption, in case any other residential or university building consumes it simultaneously (the peak load is 6.6 MWh and the consumption is 18.11 GWh/y);
- **VSC3: VSC + Decentralized prosumers –** In this scenario, ***all buildings (university and residential) act as prosumers*** that own the PV plant on their rooftop (the nominal power is 3.7 MW and the production is 5.33 GWh/y), the electricity produced is first self-consumed by each building, then sold to the grid and accounted

Table 4 –
Characterization of use-types adopted in the simulation.

	Ventilation rate	*Relative humidity*	*Sensible heat gains*	*Electric gains*		*Occupancy density*
				Appliances	*Lighting*	
Use type	***[l/p]***	***[%]***	***[W/p]***	***[W/m^2]***	***[W/m^2]***	***[m^2/p]***
Residential	5	30–70	65	8	10	30
University	7	30–60	70	8	12	10
Office	7	30–70	65	15	10	8
Laboratory	12	30–70	75	52	12	15
Library	8	30–70	60	10	12	5
Server Room	36	30–50	0	200	5	0
Reference norm	UNI EN 16798 UNI 10339	UNI EN 16798	UNI EN 16798 UNI/TS 11300	UNI EN 15193 UNI/TS 11300 UNI/TR 11552		UNI/TR 11552

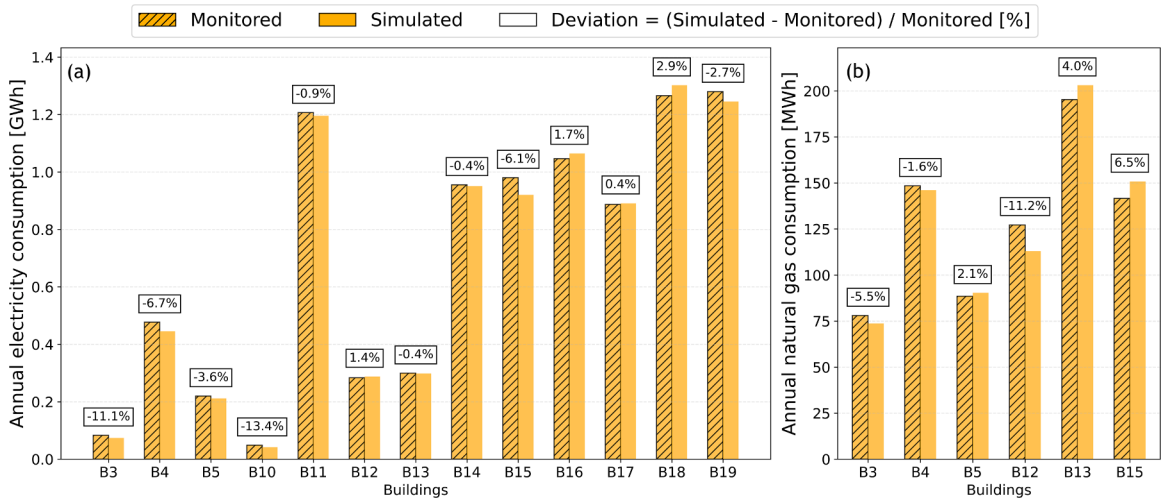


Fig. 4. Deviation between monitored and simulated consumption of university buildings.

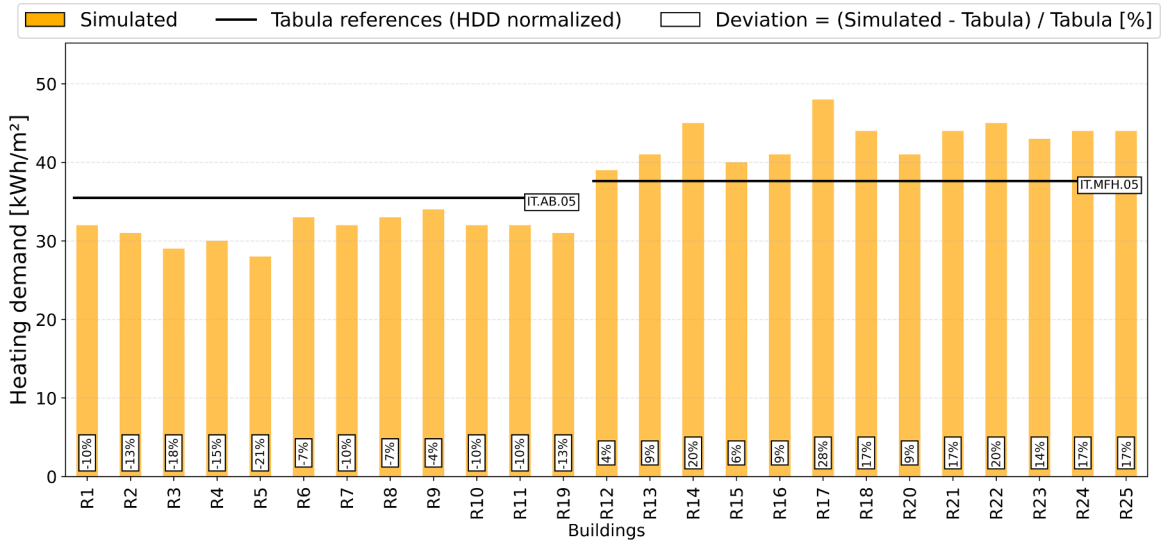


Fig. 5. Deviation between the simulated heating demand of residential buildings. and the corresponding Tabula references.

for in the calculation of the incentivized virtual self-consumption, if any other building consumes it simultaneously (same load as VSC2);

- **VNM1: VNM + Centralized producer –** In this scenario, the ***university acts as an external producer*** that provides its rooftop for PV panels (same size as VSC1), the electricity produced is allocated to each residential building using a static key that is proportional to their surface area and accounted as self-consumption (same load as VSC1), the surplus production uncovered by the allocation is sold to the grid;
- **VNM2: VNM + Centralized prosumer –** In this scenario, the ***university acts as a prosumer*** that owns the PV plant on the rooftop of its buildings (same size as VSC2), the electricity produced is allocated to each building (university and residential) using a static key proportional to their surface area and accounted as self-consumption (same load as VSC2), the surplus uncovered by the allocation is sold to the grid;
- **VNM3: VNM + Decentralized prosumers –** In this scenario, ***all buildings (university and residential) act as prosumers*** that own the PV plant on their rooftop (same size as VSC3), the electricity produced is allocated to each building (university and residential) using a static key proportional to their surface area and accounted as self-consumption (same load as VSC3), the uncovered surplus is sold to the grid.

3.4. Data analysis

3.4.1. Energy performance indicators

The energy balance for the VNM scheme starts by assigning an hourly quota of the total PV production in the community to each building according to its surface area, as outlined by Eq. 1a, where: $PV_{Production,b,h}$ and $PV_{Quota,b,h}$ are, respectively, the amount of PV electricity produced by and assigned to building *b* at the hour *h*, expressed in kWh, and A_b is the gross floor area of building *b*, expressed in m^2. For the VSC scheme, the PV production is not allocated ex-ante, so the quota is equal to the building's production, as outlined by Eq. 1b.

$$PV_{Quota\ VNM,b,h} = [\ \sum(PV_{Production,b,h}) \cdot A_b] \ / \ \sum(A_b) \quad (1a)$$

$$PV_{Quota\ VSC,b,h} = PV_{Production,b,h} \quad (1b)$$

The next step consists of calculating the hourly direct PV self-consumption ($PV_{Self\ direct,b,h}$) of each building, as shown by Eq. 2, where $E_{consumption,b,h}$ is the electricity consumed by building *b* at hour *h*, expressed in kWh. Only for the VNM scheme, when a building produces

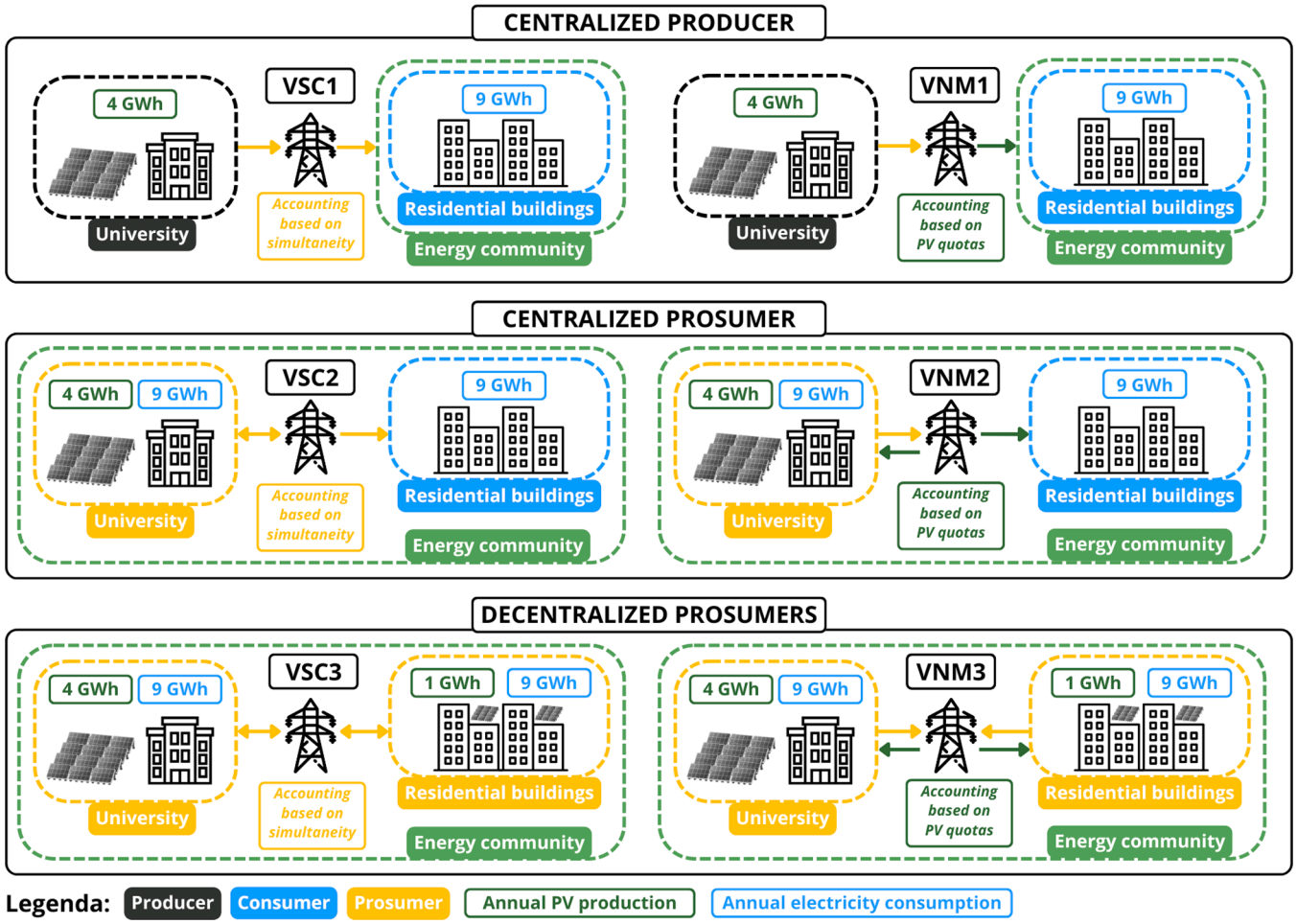


Fig. 6. Graphical illustration of the six (6) scenarios elaborated for the REC modelling.

more PV electricity than what it is assigned, the quota influences the amount of direct self-consumption. Virtual self-consumption ($PV_{Self\ virtual,b,h}$) is calculated as the absolute value of the minimum difference between PV production and either the consumption or the PV quota assigned to the building, as shown by Eq. 3. For the VNM scheme, buildings with a quota higher than their PV production account for virtual self-consumption as long as they consume enough electricity to cover it. On the other hand, virtual self-consumption is always zero for the VSC scheme.

$$PV_{Self\ direct,b,h} = min(PV_{Production}, PV_{Quota}, E_{Consumption})_{b,h} \quad (2)$$

$$PV_{Self\ virtual,b,h} = max\{0, [\ min(PV_{Quota}, E_{Consumption})_{b,h} - (PV_{Production})_{b,h}\ ]\} \quad (3)$$

The PV production surplus ($PV_{Surplus}$) and the uncovered load ($E_{uncovered}$) of each building are calculated hourly as the absolute differences between the total self-consumption and, respectively, the PV production and the electricity consumption, as shown by Eqs. 4–5.

$$PV_{Surplus,b,h} = \max[0, (PV_{Production} - PV_{Self\ direct} - PV_{Self\ virtual})_{b,h}] \quad (4)$$

$$E_{Uncovered,b,h} = max[0, (E_{Consumption} - PV_{Self\ direct} - PV_{Self\ virtual})_{b,h}\ ] \quad (5)$$

For the VSC scheme, the amount of incentivized collective virtual self-consumption (REC_{Self}) is calculated as the minimum between the total PV surplus of the whole REC and the total undercovered load within the REC, as outlined by Eq. 6.

$$REC_{Self\ VSC,h} = min[\ \sum(PV_{Surplus,b}), \sum(E_{Uncovered,b})]_h \quad (6)$$

The energy performance is evaluated using four (4) indicators, relative to direct and virtual **self-consumption rates** (DSC_{rate} and VSC_{rate}), measuring the share of PV production used by REC members, calculated using Eq. 7-8, and the direct and virtual **self-sufficiency rates** (DSS_{rate} and VSS_{rate}), calculated using Eq. 9-10. Equations 8 and 10 specify the different calculations made for VSC and VNM schemes.

$$DSC_{rate} = PV_{Self\ direct} / PV_{Production} \quad (7)$$

$$VSC_{rate,\ VSC} = (PV_{Self\ direct} + REC_{Self}) / PV_{Production} \quad (8a)$$

$$VSC_{rate,\ VNM} = (PV_{Self\ direct} + PV_{Self\ virtual}) / PV_{Production} \quad (8b)$$

$$DSS_{rate} = PV_{Self\ direct} / E_{Consumption} \quad (9)$$

$$VSS_{rate,\ VSC} = (PV_{Self\ direct} + REC_{Self}) / E_{Consumption} \quad (10a)$$

$$VSS_{rate,\ VNM} = (PV_{Self\ direct} + PV_{Self\ virtual}) / E_{Consumption} \quad (10b)$$

3.4.2. Economic analysis

The economic analysis relies on the calculation of the revenue streams enabled by the REC configurations: the savings generated by self-consumption, the sale of PV surplus electricity to the national grid, and the incentive tariff for virtual self-consumption under VSC scenarios. As outlined by Eq. 11, the electricity savings due to self-consumption are calculated hourly by multiplying the amount of PV self-consumption of the building by the cost of electricity in Italy ($c_{EL,buy}$), set as the average of all fixed price offers listed in the protected

market (equal to 300 €/MWh in 2025) (Autorità di Regolazione per Energia Reti e Ambiente ARERA, 2025). Virtual self-consumption is null for VSC scenarios, and only direct self-consumption is accounted for savings.

$$Saving\ self_{\ b,h} = (PV_{Self\ direct} + PV_{Self\ virtual})_{b,h} \cdot c_{EL,buy} \quad (11)$$

As outlined by Eq. 12, the revenues generated by selling the PV electricity surplus to the national grid were calculated by multiplying the amount of PV surplus by the hourly zonal electricity price (p_{zone}) corresponding to Sicily (the average in 2025 is equal to 116 €/MWh) (Gestore Mercati Energetici GME, 2026). A minimum sale price was also considered ($c_{EL,\ sell,\ min}$), according to the guarantees given by the Italian responsible party ARERA (equal to 46.8 €/MWh in 2024).

$$Surplus_{\ b,h} = PV_{Surplus,b,h} \cdot max(p_{zone,h}\ ,\ c_{EL,sell,min}) \quad (12)$$

As outlined by Eq. 13, the incentive tariff generated by the virtual self-consumption was calculated by multiplying the amount of electricity accounted as virtual self-consumption by the three (3) components of the tariff defined by the Italian regulations: a dynamic Premium Tariff (*TIP*), a regional correction factor (TIP_{reg}), and an additional fixed contribution given by ARERA. As shown by Eq. 14, the *TIP* is calculated based on zonal price signals, with a maximum constraint of 120 €/MWh for small-size PV plants of less than 200 kW. The regional correction factor compensates for the differences in PV yields and corresponds to an extra 10 and 4 €/MWh in the Northern and Central regions, respectively. Lastly, the ARERA contribution is equal to 10.57 €/MWh (Gestore Servizi Energetici GSE., 2025).

$$REC\ incentive_{\ h} = REC_{Self\ VSC,h} \cdot (TIP_h + TIP_{reg} + ARERA_{contribution}) \quad (13)$$

$$TIP_h = min(120,\ \{80 + max[0,\ (180 - p_{zone,h})]\})\ €/MWh \quad (14)$$

The overall revenues generated by the REC configurations are calculated as a sum of the three (3) components mentioned above. Since the objective of the analysis is to compare the effect of applying different energy sharing schemes to the same technical configurations, no economic consideration is made on the investment costs.

3.4.3. Sensitivity analysis

A sensitivity analysis is conducted to assess the influence of two key parameters: the zonal electricity price and the installed PV capacity. The input datasets for zonal prices adopted in the analysis consist of seven (7) separate years (2019–2025) and two (2) zones (Sicily and Italy) (Gestore Mercati Energetici GME, 2026). Table 5 presents the yearly average value of the main economic parameters employed for the seven years assessed. This analysis evaluates how changes in the distribution of energy costs affect REC revenues. Simultaneously, the PV capacity installed is varied in the simulation environment from a minimum of 30% exploitation of the available rooftop area up to 70%. The baseline conditions for the previous analysis correspond to the use of zonal price for Sicily in 2025 and the assumption of 50% of available rooftop area for PV production. The analysis examines the impact of PV sizing on the energy balance of each building separately and within the REC. The combined analysis reveals insights on the suitability of the incentive schemes proposed under different design and economic conditions.

Table 5
Yearly averages of the economic parameters employed in the sensitivity analysis.

	Zonal price (Italy)	*Zonal price (Sicily)*	*TIP*	*Market price*	*Guaranteed price*
2019	52 €/MWh	63 €/MWh	130 €/MWh	228 €/MWh	39.8 €/MWh
2020	38 €/MWh	46 €/MWh	131 €/MWh	189 €/MWh	40.0 €/MWh
2021	125 €/MWh	129 €/MWh	119 €/MWh	207 €/MWh	39.9 €/MWh
2022	302 €/MWh	295 €/MWh	94 €/MWh	445 €/MWh	40.7 €/MWh
2023	127 €/MWh	126 €/MWh	123 €/MWh	360 €/MWh	44.0 €/MWh
2024	108 €/MWh	112 €/MWh	127 €/MWh	234 €/MWh	46.4 €/MWh
2025	119 €/MWh	116 €/MWh	127 €/MWh	300 €/MWh	46.8 €/MWh

4. Results

4.1. Energy and economic performance

The UBEM simulations described in Section 3.2 provide the building-level electricity consumption and PV production profiles used to assess the six REC scenarios defined in Section 3.3. Fig. 7 summarises the resulting yearly electricity balances for each simulated building under the six (6) scenarios analysed, distinguishing between the electricity self-consumed (directly and virtually) and that withdrawn from the grid. The building consumption profiles of corresponding VSC and VNM scenarios are identical. The differences in the allocation of PV production between the two sharing schemes can be appreciated in Fig. 7. For example, as explained in Section 3.2, in scenarios VSC1 and VNM1 (centralized producer), university buildings act as producers, and the electricity is consumed only by the residential buildings. In scenario VNM1, a quota of the PV production (in green) is allocated to residential buildings and accounted for as virtual self-consumption (in red). Instead, in VSC1, virtual self-consumption is accounted for ex-post based on the overall REC consumption and the total PV surplus available, so it is not visible at the individual building level in Fig. 7.

Moving to centralized prosumer scenarios (VSC2 and VNM2), the same PV production is maintained, and the consumption of university buildings is also considered. For VSC2, the university buildings first benefit from direct self-consumption (in blue), then generate PV surplus that is accounted for in the calculation of virtual self-consumption. On the other hand, for VNM2, all buildings get assigned a quota of the total PV production proportional to their surface area, and can use this for self-consumption. Some university buildings (B5, B12, B16, etc.) can only benefit partly from their own PV production; thereby, their direct self-consumption gets reduced. Other university buildings (B14, B15) receive a quota higher than their own PV production; thereby, they benefit from virtual self-consumption (in red) on top of their direct self-consumption (in blue). Finally, moving to all prosumer scenarios (VSC3 and VNM3), the PV production on top of residential buildings is also considered while the same overall consumption is maintained. Again, for VNM3, all buildings get assigned a quota that can be higher or lower than their own PV production. However, since university buildings generally show a higher PV potential, most residential buildings also benefit from virtual self-consumption in VNM3. Fig. 8 compares scenarios VSC2 and VNM2, showing hourly aggregated profiles of production and consumption for an average week in winter. The total electricity consumption (in yellow) and PV production (sum of self-consumption and surplus) are the same for the two scenarios. The total amount of direct self-consumption (in blue) for VSC2 is higher than for VNM2. Similarly, the virtual self-consumption in VSC2 is higher than in VNM2, since the accounting is made ex-post and does not depend on the sharing key chosen (in this case a static key proportional to the surface area). Clearly, the application of dynamic sharing keys with more sophisticated allocation criteria (based on historical consumption or forecasting) could improve the performance of VNM schemes.

Fig. 9 and Table 6 compare the aggregated yearly energy balance results obtained for the six (6) scenarios analysed at baseline conditions (Zonal price for Sicily in 2025; 50% of available rooftop area for PV production). In VNM scenarios, the amount of virtual self-consumption assigned to buildings with a PV production lower than the quota compensates for the decrease in direct self-consumption of those buildings exceeding the quota. Still, not all the electricity quota is actually



Fig. 7. Annual consumption (left), divided into direct self-consumption (blue), allocated virtual self-consumption (red – only for VNM), and uncovered load (yellow), and PV production of university (B3-B19) and residential buildings (R1-R25). under the six (6) scenarios analysed; each bar represents one building.

consumed by the buildings, resulting in a lower self-consumption share than the corresponding VSC scenarios. Indeed, the amount of energy accounted as self-consumption (direct and virtual) is higher for VSC scenarios in all configurations, therefore all energy indicators are more favourable for VSC. Compared with centralized producer scenarios (VSC1 and VNM1), where the PV production cannot be consumed by the university, the centralized prosumer configurations (VSC2 and VNM2) show higher self-consumption shares, at the expense of self-sufficiency.

Fig. 10 and Table 7 compare the economic results obtained for the six (6) scenarios at baseline conditions (Zonal price: Sicily 2025; 50% PV potential). Direct energy savings are the most prominent revenue stream, accounting for 74–81% of total revenues in VSC scenarios and 46–69% in VNM scenarios. Direct energy savings are higher for VSC scenarios, since the amount of direct self-consumption for VNM scenarios is reduced for buildings that produce above the assigned quota, as explained above. However, in VNM scenarios, virtual energy savings are accounted at market price (300 €/MWh on average in 2025), which is much higher than both the sale price at which PV surplus is sold to the grid (equal to the zonal price) and than the incentive tariff (both equal to 116 €/MWh on average in 2025). For this reason, in centralized producer scenarios (VSC1 vs. VNM1) where there is no direct self-consumption, VNM1 is more profitable, since the electricity shared is accounted as savings. In the other cases (VSC2 vs. VNM2; VSC3 vs. VNM3), the total revenue is comparable for the two sharing schemes (around 2% differences), since the lower amount of virtual energy sharing accounted for in VNM scenarios is compensated at the more profitable market price. Lastly, regardless of the sharing scheme, centralized prosumer scenarios (VSC2 and VNM2) are more profitable than centralized producer ones (VSC1 andVNM1).

4.2. Sensitivity analysis

Fig. 11 and Fig. 12 show the results of the sensitivity analysis on energy and economic indicators by varying the amount of PV installed on buildings from 30% to 70% of the available rooftop area. Focusing on energy indicators, VSC scenarios perform better than VNM scenarios regardless of the PV power installed. Compared to direct self-consumption in VSC scenarios (blue line), which represents the baseline without energy sharing, virtual self-consumption in VNM scenarios (dotted red line) increases by 8–16% for low PV shares and by 10–25% for high PV shares. Comparatively, virtual self-consumption in VSC scenarios (red line) increases by 15–20% for low PV shares (reaching 100%) and by 20–35% for high PV shares. The energy benefit of virtual self-consumption is more prominent for high PV shares under centralized prosumer scenarios (VSC2 and VNM2).

On the economic side, the revenue obtained by VNM scenarios is comparable to or even higher than for VSC scenarios. Compared to the baseline without energy sharing (savings from direct self-consumption

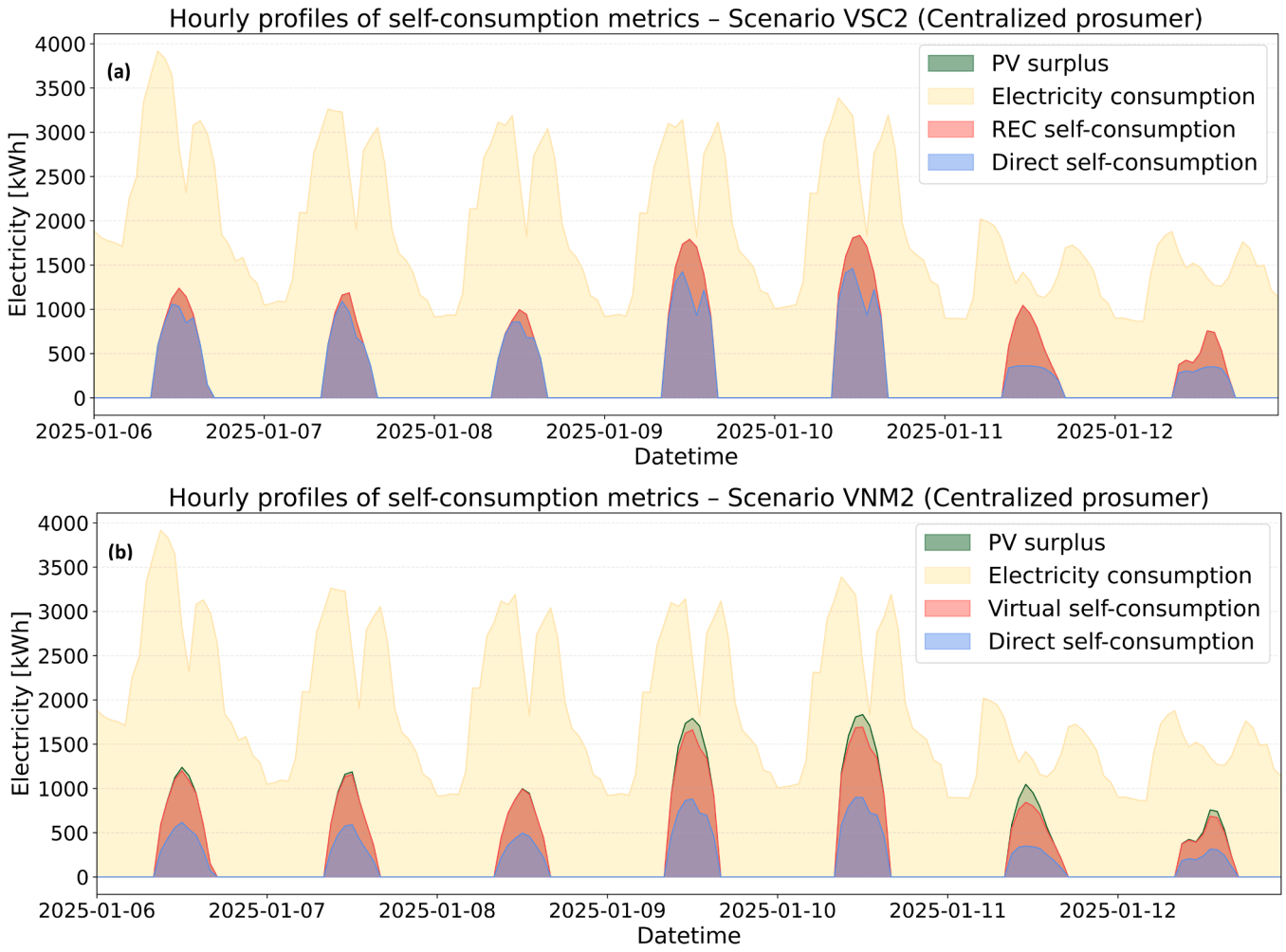


Fig. 8. Hourly profiles of direct (blue), virtual self-consumption (red), and surplus production (green) for an example week, comparing energy sharing schemes. under the same PV ownership scheme (VNM2 vs. VSC2).

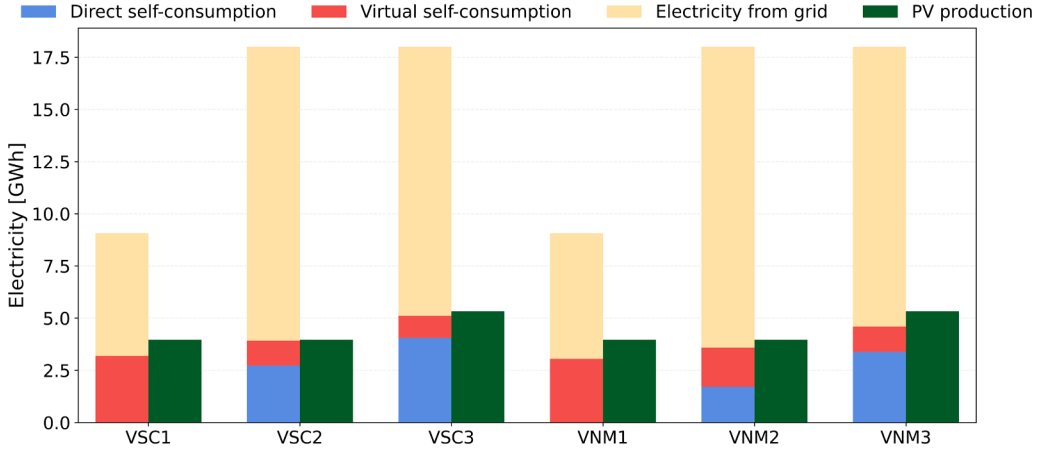


Fig. 9. Yearly energy balance under different scenarios at baseline conditions.

and sale of surplus), energy sharing in VNM scenarios increases revenues by 6–11% for low PV shares and by 8–21% for high PV shares. Comparatively, in VSC scenarios, revenues increase by 7–9% for low PV shares and by 10–20% for high PV shares. Again, the economic benefits of energy sharing are more prominent for high PV shares in centralized prosumer scenarios (VSC2 and VNM2). With the increase in PV power, the relative revenues with energy sharing (black lines) decrease at a slower pace than the baseline (green line). For instance, by doubling the PV share from 30% to 60%, the relative revenues decrease by 4–6% for VSC and by 5–7% for VNM. Comparatively, the relative revenues in the baseline (obtained by direct self-consumption and the sale of surplus) decrease by 11%. The stabilization of relative revenues for higher PV shares might incentivize investments in larger projects.

As shown in Fig. 13, another sensitivity analysis was obtained by

Table 6
Energy balance indicators under different scenarios at baseline conditions.

VSC – Virtual Self-Consumption	***DSC_{rate}***	***VSC_{rate}***	***DSS_{rate}***	***VSS_{rate}***
VSC1: *Centralized producer*	-	80.5%	-	35.2%
VSC2: *Centralized prosumer*	68.8%	99.0%	15.1%	21.8%
VSC3: *Decentralized prosumers*	76.0%	95.9%	22.5%	28.4%
VNM – Virtual Net-Metering	***DSC_{rate}***	***VSC_{rate}***	***DSS_{rate}***	***VSS_{rate}***
VNM1: *Centralized producer*	-	76.8%	-	33.6%
VNM2: *Centralized prosumer*	43.2%	90.5%	9.5%	19.9%
VNM3: *Decentralized prosumers*	63.6%	86.2%	18.8%	25.5%

varying the price signals with historic values from 2019 to 2025 for the zone corresponding to Sicily, as reported in Table 5. Generally, the amount of revenue is directly linked to market price signals, since both direct self-consumption and virtual self-consumption in VNM scenarios are accounted for as bill savings. Therefore, results do not show significant differences between VSC and VNM scenarios, except when the university acts as an external producer (VSC1 vs. VNM1). In this case, where direct self-consumption is null, VSC1 strongly relies on the revenues generated by surplus production sold at the zonal price. Regardless of the sharing scheme adopted, it is always more profitable for the university to act as a prosumer instead of an external producer (VSC1 vs. VSC2; VNM1 vs. VNM2), with the only exception of 2021.

5. Discussion

The results address the two main dimensions investigated in this study and show that the ownership structure of the generation assets and the sharing scheme adopted by a mixed-use REC have measurable impacts on its energy and economic performance. Focusing on energy sharing, VSC schemes consistently achieve higher self-consumption and self-sufficiency rates than VNM schemes across all scenarios. The ex-post accounting mechanism of VSC schemes allows a flexible matching between production and demand, and the full utilisation of PV surplus. On the contrary, the ex-ante allocation adopted by VNM schemes based on a static sharing key reduces the amount of direct self-consumption of buildings producing above their assigned quota, potentially leading to a misalignment between assigned generation and consumption. At the same time, the economic outcomes of the two sharing mechanisms remain broadly comparable due to the different ways shared electricity is valued. While VSC schemes benefit from higher direct self-consumption, VNM schemes can compensate through higher revenues from virtual energy sharing, since shared electricity is accounted for as bill savings at market price. Significant differences only emerge in centralized producer models (VSC1 and VNM1), where VNM is more profitable because of the lack of direct self-consumption. Sensitivity analyses further highlight the better energy performance of VSC schemes across different PV power levels, and especially for high PV shares. For both schemes, energy sharing helps stabilize the relative revenues as PV power installed increases, potentially supporting the economic viability of larger PV installations. Still, the economic benefits of energy sharing are strongly influenced by market price variations, with profitability increasing in periods of high electricity prices. Results do not show significant differences between VSC and VNM schemes, except for the centralized producer model (VSC1 and VNM1). Regarding ownership, configurations with a large centralized prosumer (VSC2 and VNM2), in this case the university, achieve the best energy indicators. The economic benefits of energy sharing are also more prominent in these configurations, compared to individual self-consumption. On the contrary, the centralized producer model (VSC1 and VNM1) shows the lowest levels of direct self-consumption, as it relies mainly on the allocation of surplus, making it the least favourable configuration from an energy and economic perspective. These results indicate that non-residential buildings contribute to REC performance through their generation potential and the interaction between their production and consumption profiles and those of other members. For large PV producers, participating in the REC as prosumers leads to higher overall benefits compared to acting as external producers, particularly if nearby residential users have limited installation potential or investment capacity. From a collective perspective, energy sharing increases the installed PV capacity without a proportional reduction in economic viability, as the additional generation can be redistributed within the community rather than exported as surplus. These considerations hold for both VSC and VNM schemes, although the magnitude and distribution of revenues differ depending on the regulatory framework adopted. The implications for individual actors ultimately depend on how the revenues from shared energy are redistributed within the REC, which is not modelled in this study, but may influence the preference for centralized versus distributed prosumer configurations. These findings are particularly relevant for urban areas in the Mediterranean region,

Table 7
Economic indicators under different scenarios at baseline conditions.

VSC – Virtual Self-Consumption	***Direct savings***	***Incentive***	***Surplus***	***Revenue***
VSC1: *Centralized producer*	-	406,250 €	459,686 €	**865,936** €
VSC2: *Centralized prosumer*	817,969 €	152,266 €	143,438 €	**1113,673** €
VSC3: *Decentralized prosumers*	1216,024 €	134,581 €	148,201 €	**1498,806** €
VNM – Virtual Net-Metering	***Direct savings***	***Virtual savings***	***Surplus***	***Revenue***
VNM1: *Centralized producer*	-	913,692 €	106,429 €	**1020,121** €
VNM2: *Centralized prosumer*	513,823 €	562,123 €	43,697 €	**1119,643** €
VNM3: *Decentralized prosumers*	1016,391 €	362,865 €	85,093 €	**1464,349** €

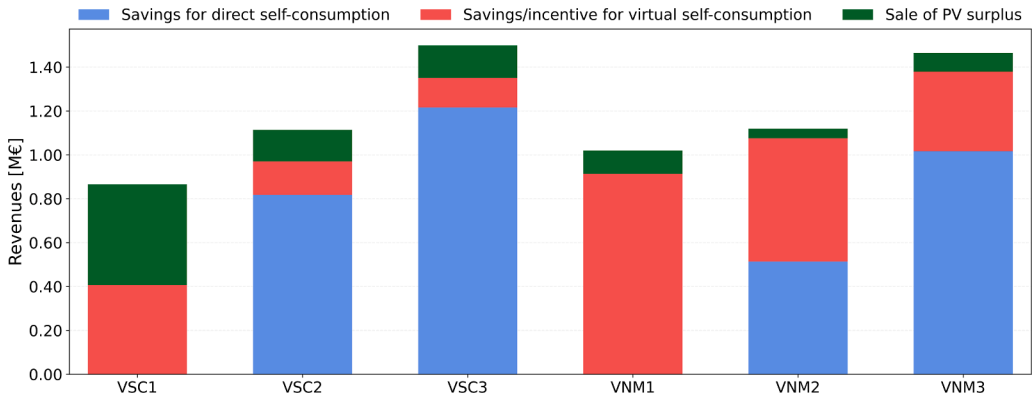


Fig. 10. Revenue streams under different scenarios at baseline conditions.

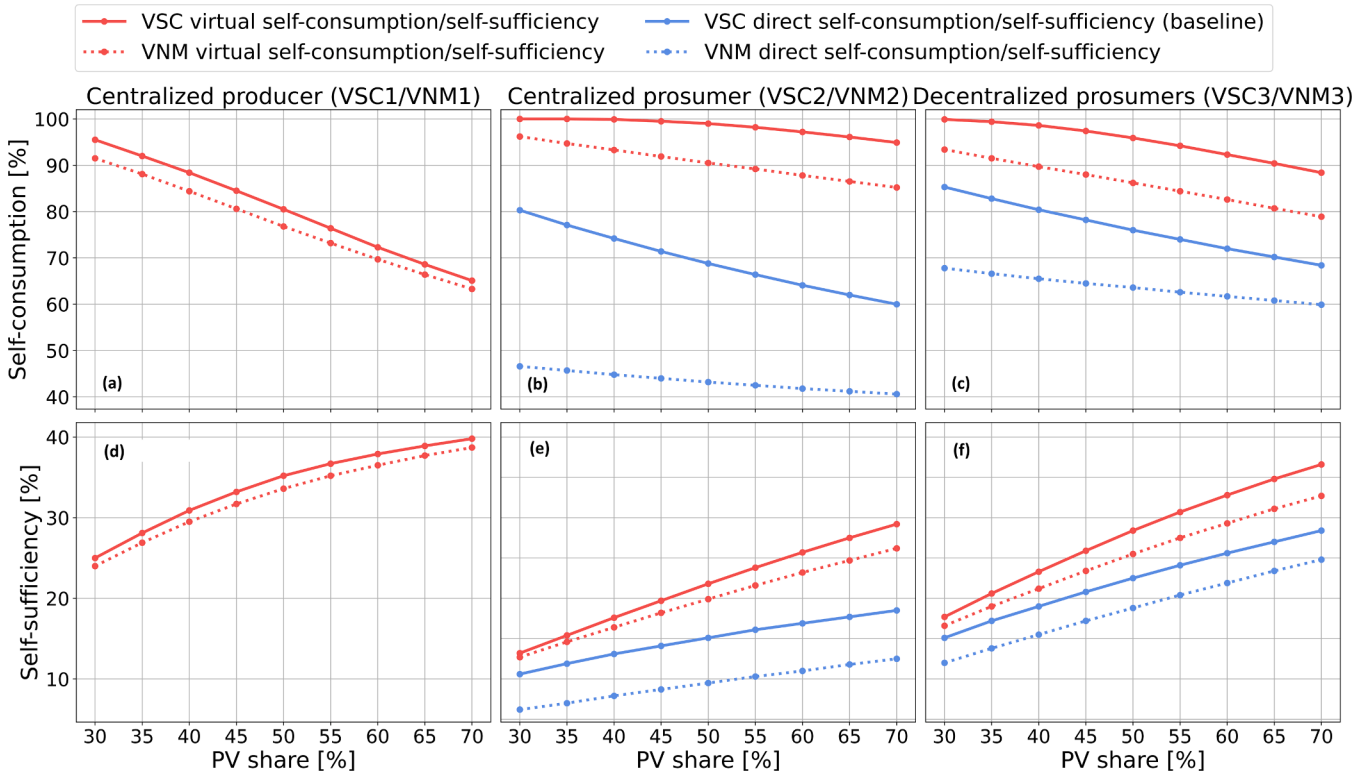


Fig. 11. Sensitivity of energy balance indicators to the PV share installed.

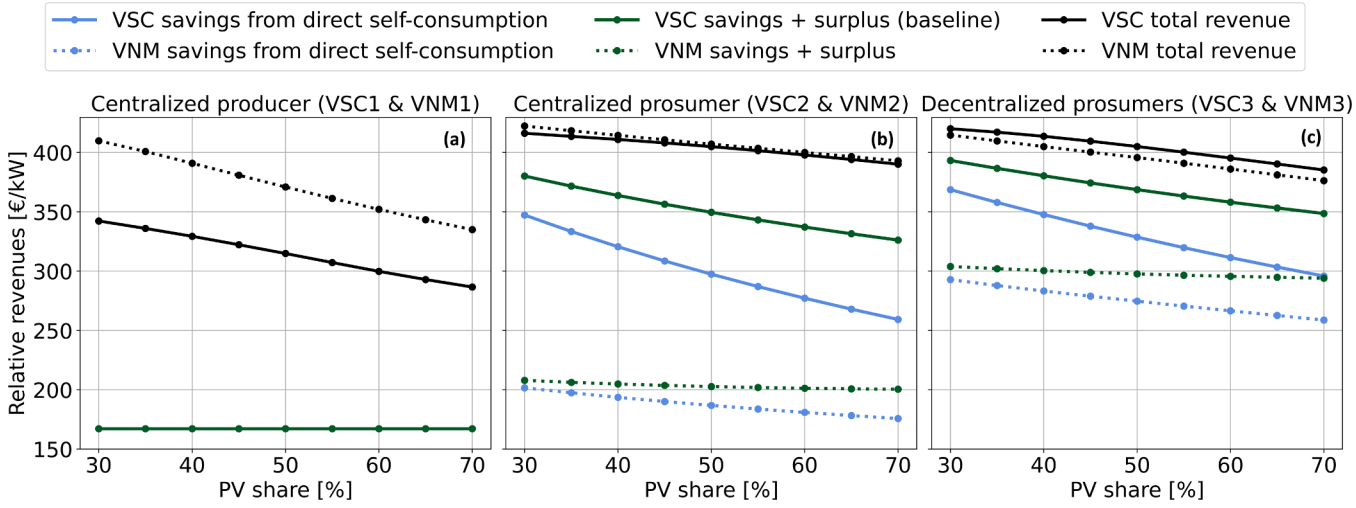


Fig. 12. Sensitivity of economic indicators to the PV share installed.

where high solar irradiance and the proximity of residential districts to large public or commercial buildings create favourable conditions for mixed-use RECs. The integration of non-residential prosumers can enhance load matching and increase the share of self-consumed energy in similar urban contexts, although the extent of these benefits remains dependent on the specific regulatory framework, particularly in relation to allocation rules and incentive structures.

6. Conclusions

This study analyses the techno-economic performance of a mixed-use Renewable Energy Community (REC), composed of a university campus and a residential district. The analysis considered six (6) scenarios representing alternative combinations of two (2) energy sharing schemes – Virtual Self-Consumption (VSC) and Virtual Net Metering (VNM) – and three (3) PV ownership configurations, ranging from a centralised model with a single institutional producer to decentralised configurations in which multiple buildings act as prosumers. Overall, the results highlight that the interaction between regulatory frameworks and ownership structures plays a critical role in shaping the energy and economic performance of RECs. In particular, VSC schemes consistently outperform VNM schemes in terms of energy performance, although the economic outcomes remain more balanced due to differences in how shared electricity is valued. Regarding ownership, mixed-use configurations involving non-residential prosumers prove more effective than centralized producer setups. In particular, enabling large PV owners to operate as prosumers enhances overall community benefits, while simultaneously allowing higher PV penetration without a significant loss in

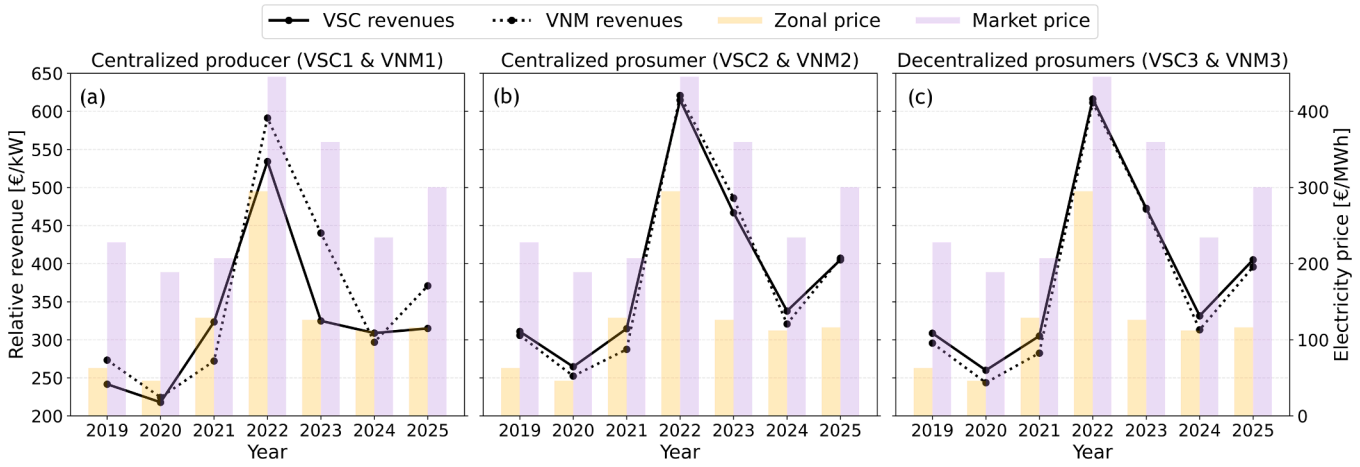


Fig. 13. Sensitivity of total REC revenues to the zonal price signals.

economic viability, thanks to the redistribution of surplus generation within the REC. Internal decisions of the REC and future policy developments could significantly influence this balance. For instance, the adoption of dynamic sharing keys could improve the load-matching performance of VNM schemes, while higher incentive levels could strengthen the economic attractiveness of VSC-based communities. At the same time, the optimization strategies for energy sharing proposed in the literature could be applied to both schemes, suggesting that the relative advantages ultimately depend on their regulatory implementation. The UBEM-based modelling framework developed in this study provides a useful tool to systematically evaluate the combined effects of energy sharing rules, ownership structures, and local energy demand profiles, supporting the design and policy assessment of mixed-use RECs involving heterogeneous actors. The results suggest that the design of mixed-use RECs should prioritise configurations in which non-residential actors operate as prosumers, as this maximises the efficiency of energy sharing through improved alignment between generation and demand profiles. However, the economic outputs are largely decoupled from energy performance and are mainly determined by accounting rules, including incentive tariffs and market price structures, which can offset or even override the advantages observed in energy indicators. Some limitations point to directions for further research. For example, the current framework does not account for more complex sharing mechanisms, such as dynamic pricing or P2P trading, nor for demand-side flexibility or storage, which could further enhance load matching, especially for high PV installation capacities.

Author contributions

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs

CRediT authorship contribution statement

Marcello Avanzini: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Francesco Guarino:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Maurizio Cellura:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.egyr.2026.109785.

Data availability

Data will be made available on request.

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